

Python Programming - 2301CS404

Lab - 9

Roll No : 459

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**01) Write a function to calculate BMI given mass and height.
(BMI = mass/h**2)**

```
In [7]: mass , h = input("Enter Mass And h : ").split(" ")
mass = float(mass)
h = float(h)

def calBMI(mass , h):
    BMI = mass / h ** 2

    return BMI

res = calBMI(mass,h)

print(res)
```

Enter Mass And h : 40 5
1.6

02) Write a function that add first n numbers.

```
In [8]: n = int(input("Enter Number n : "))

def printFirstN(n):
    add = 0
    for i in range(1,n+1):
        add += i

    return add

print(printFirstN(n))
```

Enter Number n : 5
15

03) Write a function that returns 1 if the given number is Prime or 0 otherwise.

```
In [16]: n = int(input("Enter Number : "))

def isPrime(n):
    for i in range(2,int(n**0.5)+1):
        if(n % i == 0):
            return 0

    return 1

print(isPrime(n))
```

Enter Number : 7
1

04) Write a function that returns the list of Prime numbers between given two numbers.

```
In [17]: a , b = input("Enter Range a And b : ").split(" ")
a = int(a)
b = int(b)

def isPrime(n):
    for i in range(2,int(n**0.5)+1):
        if(n % i == 0):
            return 0

    return 1

def primeRange(a,b):
    li = []

    for i in range(a,b+1):
        if(isPrime(i) == 1):
            li.append(i)

    return li

print(primeRange(a,b))
```

Enter Range a And b : 3 19
[3, 5, 7, 11, 13, 17, 19]

05) Write a function that returns True if the given string is Palindrome or False otherwise.

```
In [19]: s = input("Enter String : ")

checkPalindrome = lambda x : x == x[::-1]

print(checkPalindrome(s))
```

Enter String : oko

True

06) Write a function that returns the sum of all the elements of the list.

```
In [ ]: li = input("Enter list : ").split(" ")

li = list(map(int,li))

sumList = lambda x : sum(x)

print(sumList(li))
```

Enter list : 1 2 3 4
10

07) Write a function to calculate the sum of the first element of each tuples inside the list.

```
In [2]: li = [(1,2),(3,4),(5,6)]

def sumListTuple(li):
    s = 0
    for i in li:
        s += i[0]
    return s

print(sumListTuple(li))
```

9

08) Write a recursive function to find nth term of Fibonacci Series.

```
In [5]: n = int(input("Enter nTh Term : "))

# 0 1 1 2 3 5 8 13

def fibonacciNth(n):
    if(n == 0 or n == 1):
        return n
    return fibonacciNth(n-2) + fibonacciNth(n-1)

fibonacciNth(n-1)
```

Enter nTh Term : 5

Out[5]: 3

09) Write a function to get the name of the student based on the given rollno.

Example: Given dict1 = {101:'Ajay', 102:'Rahul', 103:'Jay', 104:'Pooja'} find name of student whose rollno = 103

```
In [7]: dict1 = {101:'Ajay', 102:'Rahul', 103:'Jay', 104:'Pooja'}

def getName(dict1,roll):
    return dict1.get(roll)

print(getName(dict1,103))
```

Jay

10) Write a function to get the sum of the scores ending with zero.

Example : scores = [200, 456, 300, 100, 234, 678]

Ans = 200 + 300 + 100 = 600

```
In [9]: scores = [200, 456, 300, 100, 234, 678]

def endWithZero(scores):
    s = 0
    for i in scores:
        if(i % 10 == 0):
            s += i
    return s

print(endWithZero(scores))
```

600

11) Write a function to invert a given Dictionary.

hint: keys to values & values to keys

Before : {'a': 10, 'b':20, 'c':30, 'd':40}

After : {10:'a', 20:'b', 30:'c', 40:'d'}

```
In [10]: di = {'a': 10, 'b':20, 'c':30, 'd':40}

def invert(di):
    return { v : k for k , v in di.items()}

print(invert(di))
```

{10: 'a', 20: 'b', 30: 'c', 40: 'd'}

12) Write a function that returns the number of uppercase and lowercase letters in the given string.

example : Input : s1 = AbcDEfgh ,Oupthput : no_upper = 3, no_lower = 5

```
In [13]: s1 = "AbcDEfgh"

def countUpperCase(s1):
    no_upper = 0
    no_lower = 0
```

```

for ch in s1:
    if(ch.isupper()):
        no_upper += 1
    else:
        no_lower += 1

return no_upper , no_lower

print(countUpperCase(s1))

```

(3, 5)

13) Write a lambda function to get smallest number from the given two numbers.

```

In [18]: a , b = input("Enter 2 Number : ").split(" ")

a = int(a)
b = int(b)

small = lambda a,b:(a,b)[b<a]

small(a,b)

```

Enter 2 Number : 4 5

Out[18]: 4

14) For the given list of names of students, extract the names having more that 7 characters. Use filter().

```

In [21]: names = ["hey" , "python" , "ok" , "okokokk" , "parisssss"]

res = list(filter(lambda x : len(x) >= 7,names))

res

```

Out[21]: ['okokokk', 'parisssss']

15) For the given list of names of students, convert the first letter of all the names into uppercase. use map().

```

In [22]: names = ["hey" , "python" , "ok" , "okokokk" , "parisssss"]

res = list(map(lambda x : x.title(),names))

res

```

Out[22]: ['Hey', 'Python', 'Ok', 'Okokokk', 'Parisssss']

16) Write udfs to call the functions with following types of arguments:

1. Positional Arguments
2. Keyword Arguments

3. Default Arguments
4. Variable Length Positional(*args) & variable length Keyword Arguments (**kwargs)
5. Keyword-Only & Positional Only Arguments

```
In [34]: def one(a,b):
          print(a,b)

          one(1,4)

          def two(**kw):
              print(kw)

          two(name="vi",age=19)

          def three(c,d=0):
              print(c,d)

          three(5)

          def four(*li , **kw):
              print("li : ",li)
              print("kw : ",kw)

          four(1,2,3,name="vi",age=19)

          def five(e,f,/,*,g,h):
              print(e,f,g,h)

          five(10,20,g=90,h=100)

1 4
{'name': 'vi', 'age': 19}
5 0
li :  (1, 2, 3)
kw :  {'name': 'vi', 'age': 19}
10 20 90 100
```

In []: